Fraud Document Detection Tool

(FDDT)

Company User Manual

A guide for client organizations: User (Submitter), Reviewer L1 and Reviewer L2

Describes the application as implemented — companion to the technical documentation in docs/

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FDDT Company User Manual

Fraud Document Detection Tool — Guide for Submitting & Reviewing Organizations

Welcome to the **Fraud Document Detection Tool (FDDT)**. This manual provides a complete, step-by-step operational guide for client organizations using FDDT to authenticate supporting documents, detect digital tampering and alterations, and streamline the review process for invoices, reimbursements, quotations, and official records.

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1. System Overview & Architecture

The **Fraud Document Detection Tool (FDDT)** is an enterprise document verification platform. When third-party supporting documents (such as vendor invoices, school fee receipts, procurement quotations, or travel receipts) are submitted, the automated verification engine performs:

1. **Intake Validation**: Checks file format (PDF), verifies integrity, ensures the file is not password-protected, and validates file size limits.
2. **Text Extraction & Normalization**: Automatically reads all printed text (including multi-lingual content such as Arabic and English) and structures key fields (dates, amounts, invoice numbers, vendor names).

3. **Forensic Analysis**: Scans for copy-move cloning, digital alteration artifacts (Error Level Analysis), text deleted or rewritten in converted scans (ghost text), retyped text in a different font or a second copy of the font, suspicious PDF metadata histories (including documents built as web pages), and visual inconsistencies.
4. **Consistency & Verification**: Validates calculations, compares cross-document values within the case, checks the vendor against your company's known **Issuer Registry**, and performs signature/stamp detection.
5. **Explainable Risk Scoring**: Aggregates findings into a transparent risk score (0–100) mapped to risk tiers (**Low**, **Medium**, **High**) with clear explanations.

> **Tenant Data Privacy**: Your organization operates in an isolated workspace. Users within your company only ever see cases, documents, and settings belonging to your company.

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2. Company User Roles & Permissions

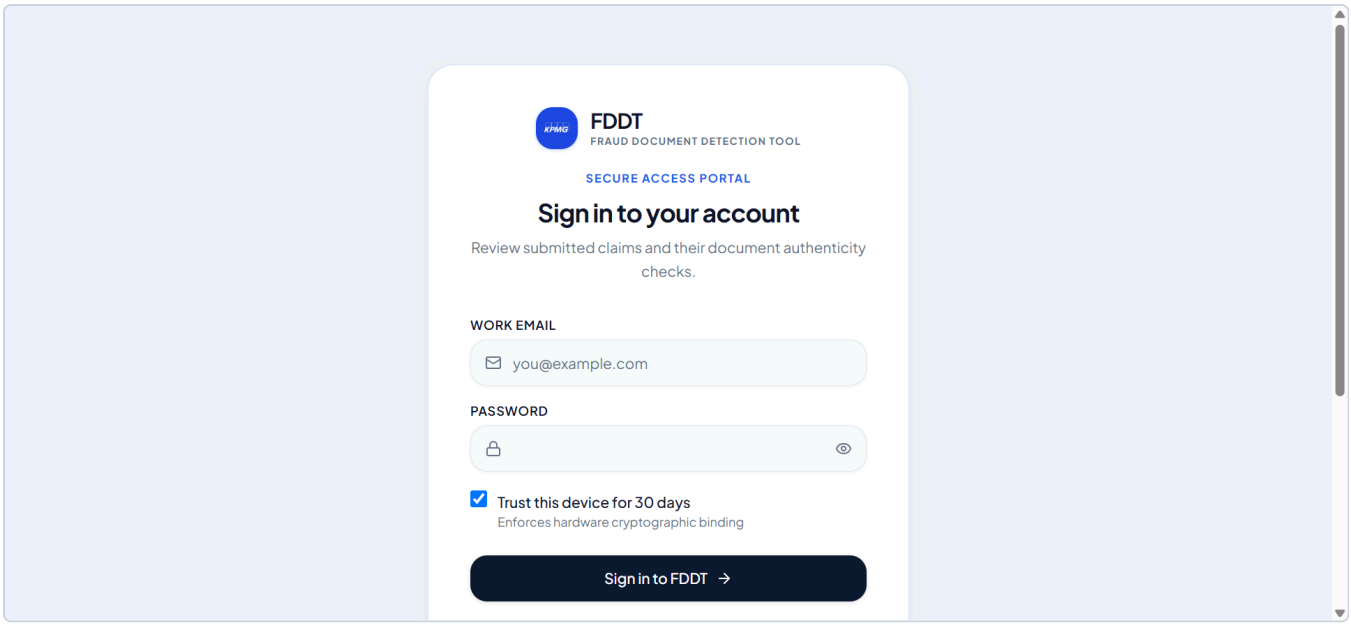
Within your company, three distinct user roles exist to enforce separation of duties:

Role	Primary Responsibility	Available Screens & Capabilities
User *(Submitter)*	Document Intake & Tracking	• **New Case**: Submit individual cases with PDF files and optional signature references. • **Bulk Upload**: Upload a zip file containing multiple case folders. • **My Cases**: Track status (*Under review*, *Cleared*, *Action required*) of own submissions. • **Case Detail (Submitter View)**: View uploaded documents and extracted fields without risk scoring details.
Reviewer L1 *(Analyst)*	Initial Verification & Risk Triage	• *All User capabilities*, plus: • **Dashboard**: High-level case volume and risk distribution metrics. • **Review Queue**: Filter cases by risk tier, document category, and status. • **Full Case Detail**: View risk scores (0–100), triggered fraud reasons, and automated forensic checks. • **Visual Overlay Viewer**: Inspect bounding box anomalies directly on document pages. • **Decision Actions**: Approve, Reject, or Escalate cases. • **Export Report**: Download full forensic audit reports.
Reviewer L2 *(Senior Reviewer)*	Final Adjudication & Company Governance	• *All Reviewer L1 capabilities*, plus: • **Escalated Cases Resolution**: Final approve/reject authority on complex cases escalated by L1 analysts. • **Issuer Registry**: Add, edit, verify, or deactivate trusted vendors and issuers for your company. • **Company Risk Rules**: Customize fraud rule weights and adjust Low/Medium/High risk tier score thresholds.

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3. Signing In & Navigation

Access the platform URL provided by your organization using your business email address and password.



FDDT Sign In Screen

1. Enter your corporate email address and assigned password.
2. Click **Sign in**.
3. Upon successful sign-in, the top navigation bar displays your organization name, user role badge, and relevant navigation links based on your permissions.

> *Note: If you have forgotten your password or require a new user account, please email your administrator for this.*

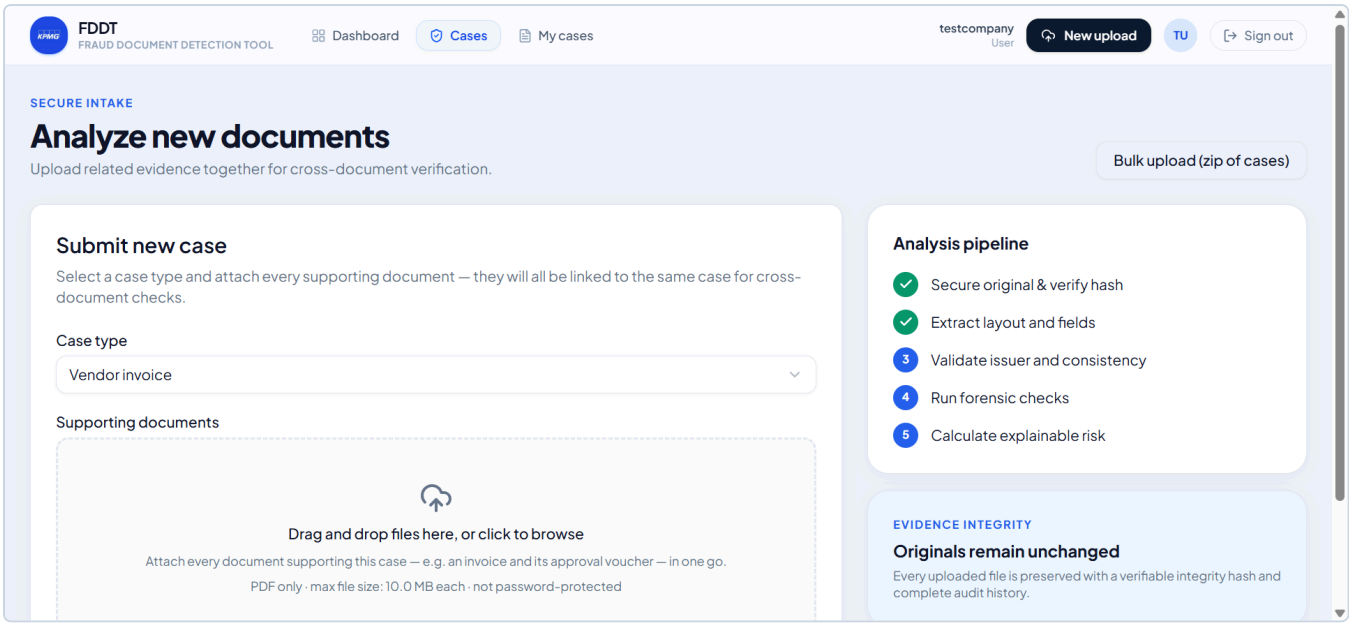
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4. User (Submitter) Guide

Submitting a Single Case

To submit documents for verification:

1. Click **New Case** (or **New upload**) in the top navigation bar.
2. Select the **Case Type** that matches your submission:
   - *School / educational document*
   - *Vendor invoice*
   - *Commercial invoice*
   - *Procurement documentation*
   - *Quotation*
   - *Travel / accommodation reimbursement*
   - *Other*

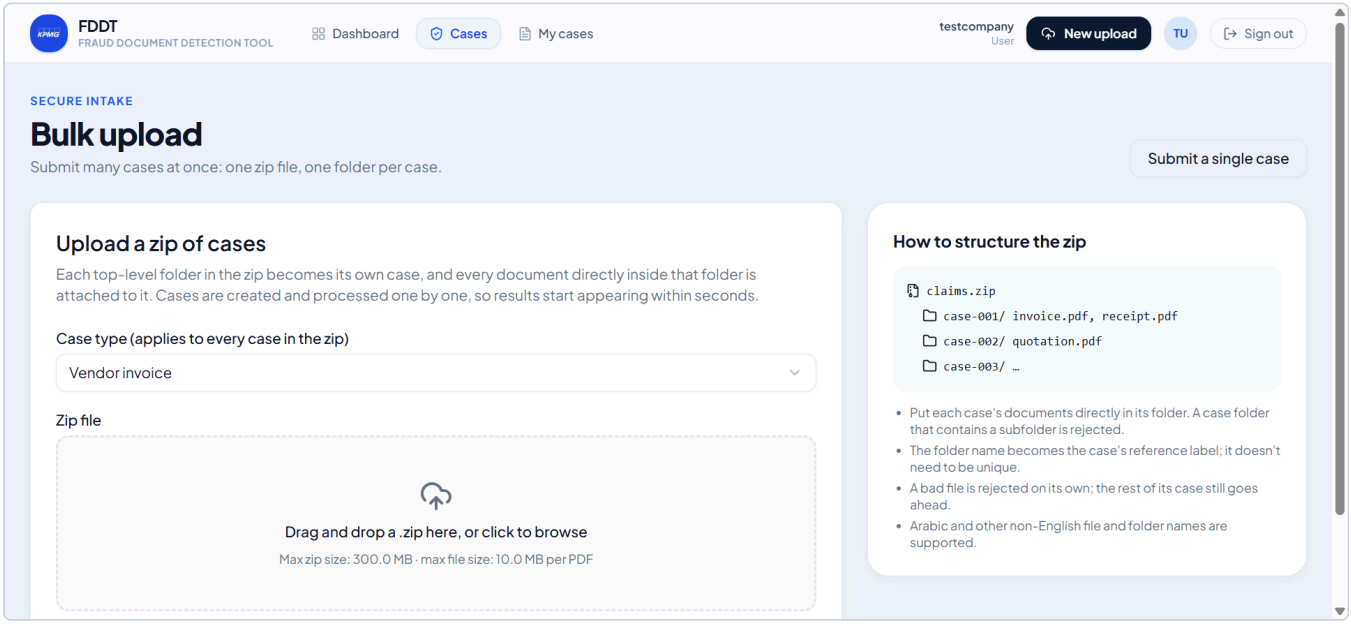


New Case Intake Form

3. **Attach PDF Files**:
   - Drag and drop your PDF documents into the upload zone or click **Browse files**.
   - **File Requirements**: Files must be valid PDF documents within your company's maximum file size limit (standard limit is 10 MB per document). Password-protected PDFs are rejected immediately for security.
4. **Optional Reference Signature**:
   - If your document contains an authorized signature or stamp that should be compared against other documents in the case, you can define a reference signature box.
5. Click **Submit Case**. The documents will begin automated processing immediately.

Submitting Cases in Bulk (Zip Upload)

When managing high volumes of invoices or expense claims, use the **Bulk Upload** feature:



Bulk Upload Interface

1. Navigate to **New Case** and select **Bulk upload (zip of cases)**.

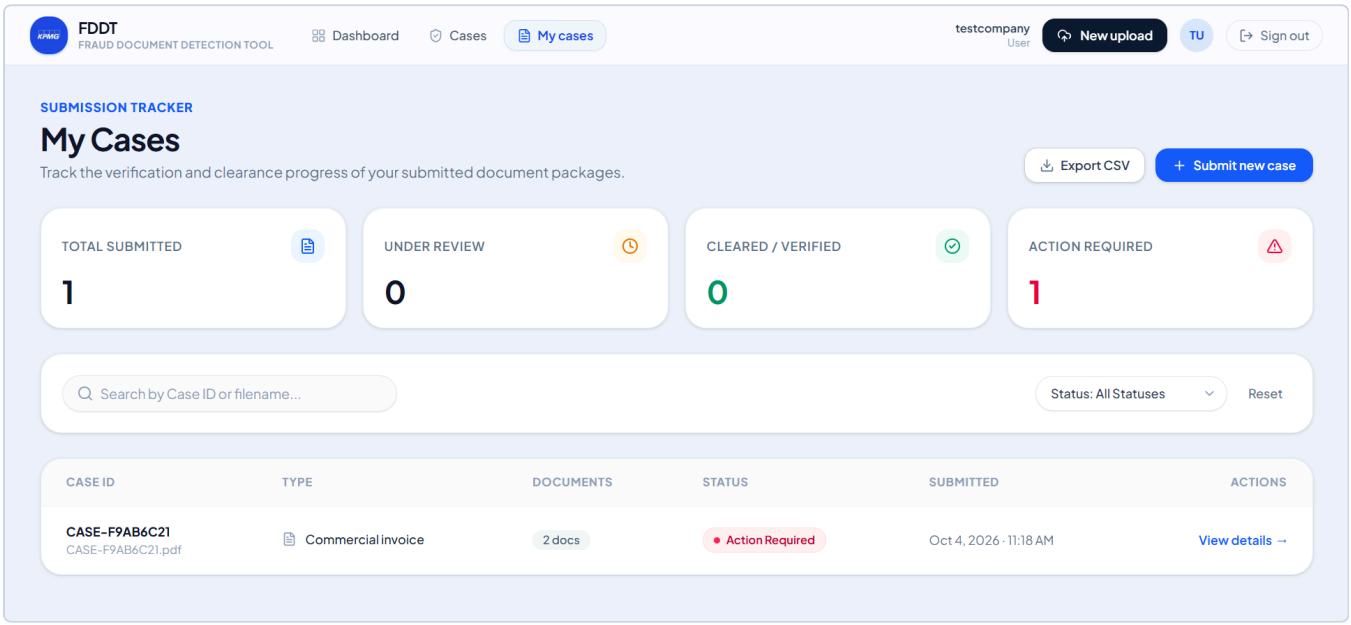
2. Prepare a standard `.zip` file on your computer structured with **one folder per case**:

```
batch_october.zip
├── CASE_INV_1001/
│   ├── invoice_1001.pdf
│   └── purchase_order.pdf
├── CASE_INV_1002/
│   └── invoice_1002.pdf
└── CASE_INV_1003/
    ├── tuition_receipt.pdf
    └── student_id.pdf
```

3. Select the default **Case Type** for the batch.
4. Drop the `.zip` archive into the intake area.
5. The system automatically extracts each folder, verifies the PDF files, and creates individual cases with automated progress indicators (**Queued** → **Processing** → **Complete**).

Tracking Submissions (My Cases)

Submitters can track all their submitted cases under **My Cases**:

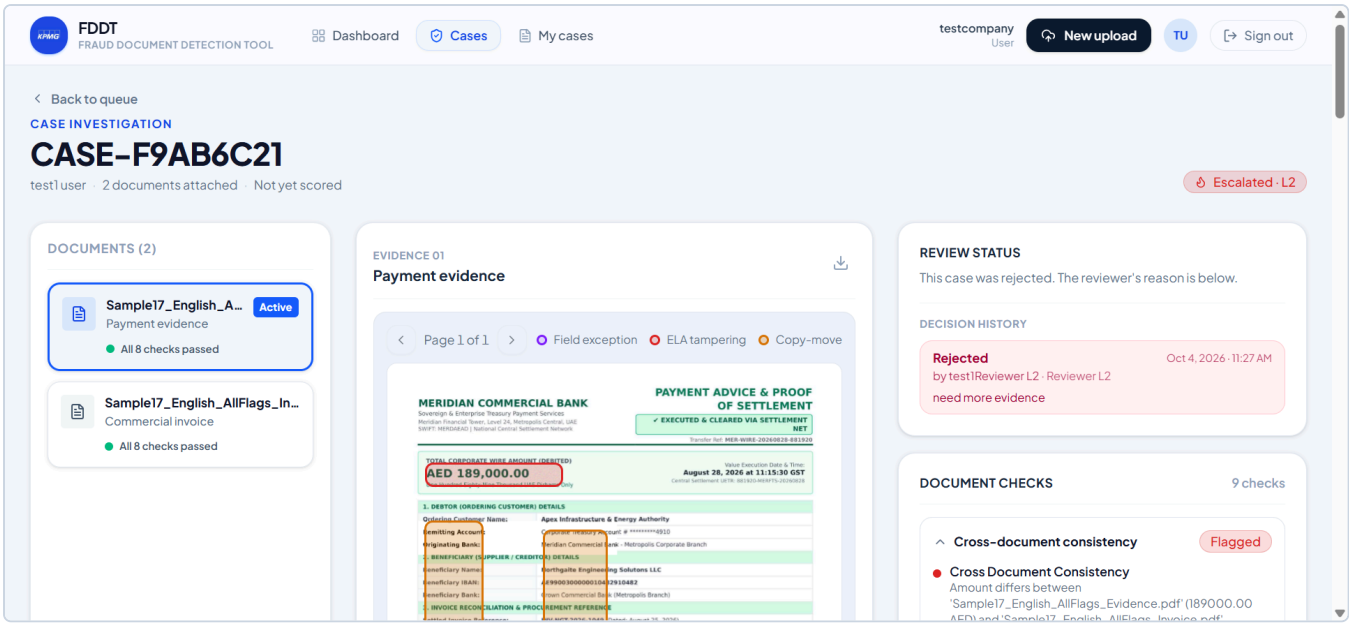


My Cases Dashboard

- **Search**: Search by case reference number or document name.
- **Filters**:
  - **All**: All cases submitted by your user account.
  - **Under review**: Cases currently being analyzed by the automated pipeline or queued for reviewer triage.
  - **Cleared**: Cases approved by the reviewing team.
  - **Action required**: Cases rejected with feedback notes from the reviewer.

Understanding the Submitter Case View

When a submitter opens one of their cases, a clean, submitter-tailored view is presented:



Submitter Case Detail View

- **Document Overview**: Lists all uploaded files, page counts, file sizes, and processing statuses.
- **Extracted Information**: Displays values recognized by the OCR engine (such as invoice number, total amount, currency, and date).
- **Activity Timeline**: Displays a chronological event log showing when the case was submitted, when files were processed, and final decisions.

> **Security & Fraud Prevention Design**: Submitter accounts do **not** display numerical risk scores, triggered fraud check rules, or forensic detection algorithms. This prevents malicious actors from probing or attempting to bypass verification thresholds.

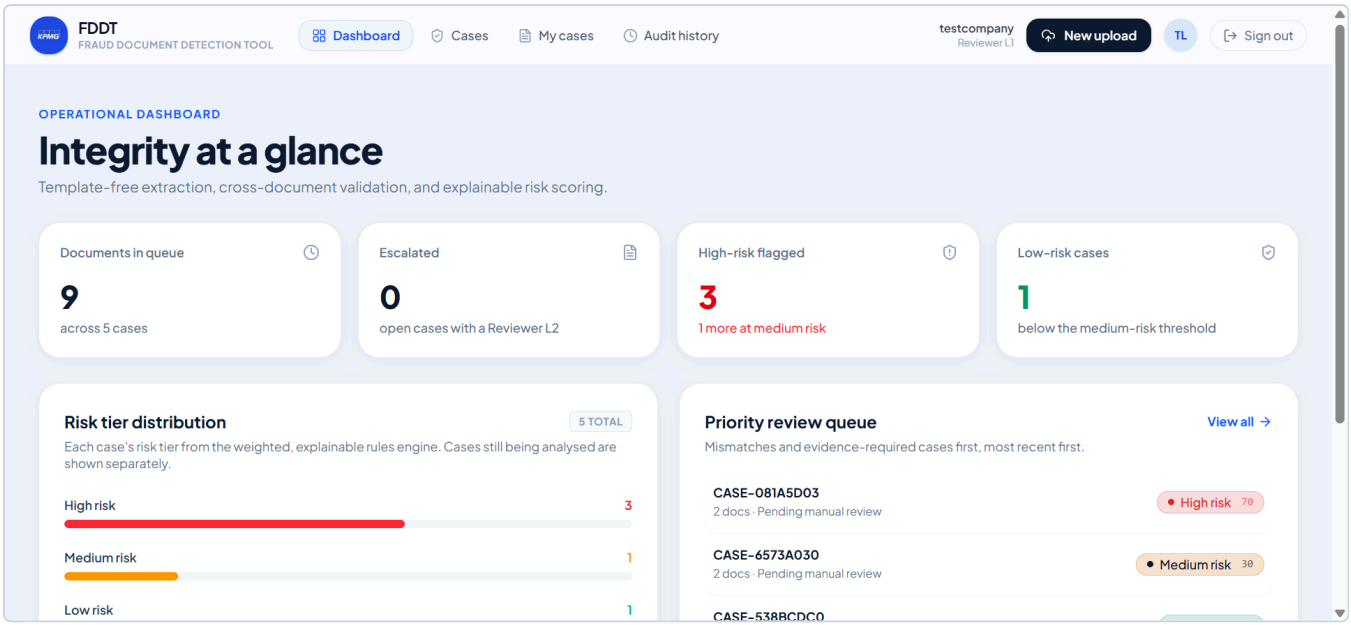
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5. Reviewer L1 (Analyst) Guide

Reviewer L1 analysts assess automated findings, inspect flagged anomalies, and make informed operational decisions.

Operational Reviewer Dashboard

The **Dashboard** provides real-time visibility into your company's verification queue:

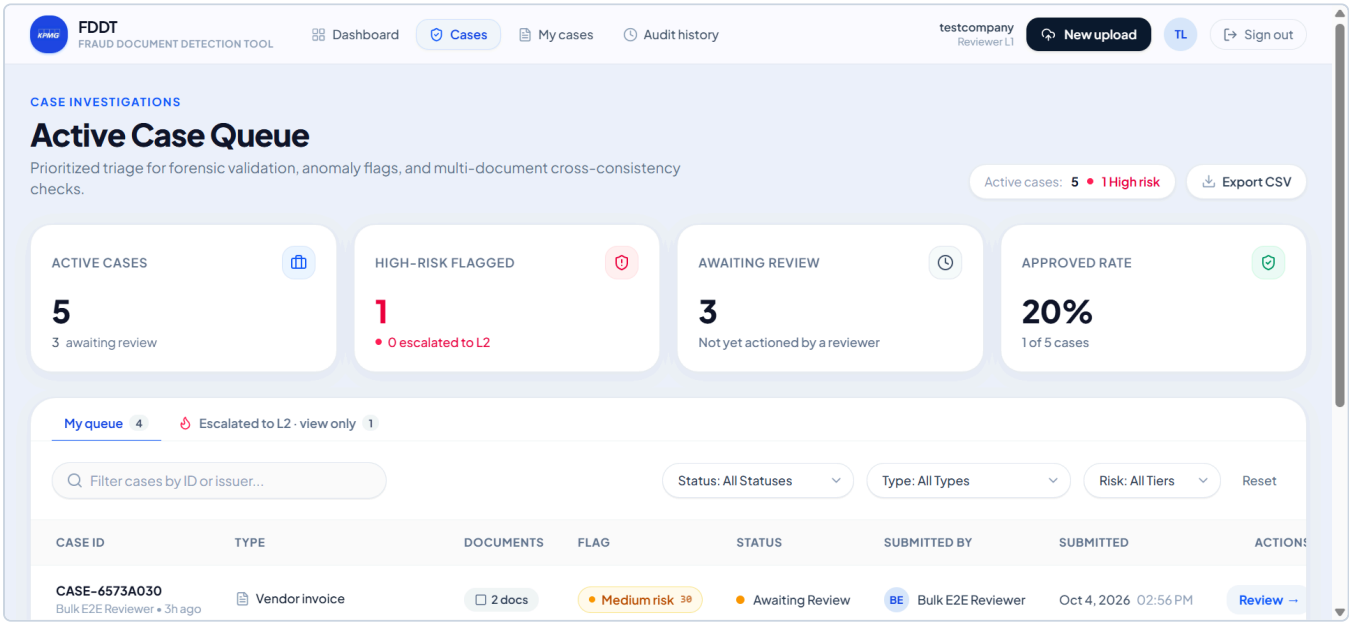


Reviewer Operational Dashboard

- **Key Metrics**: Total active cases, cases requiring immediate review, cleared cases, and flagged cases.
- **Risk Distribution**: High, Medium, and Low risk breakdowns across recent submissions.
- **Review Workload**: Cases sorted by submission date and SLA urgency.

Managing the Review Queue

Click **Cases** in the navigation bar to access the full **Review Queue**:



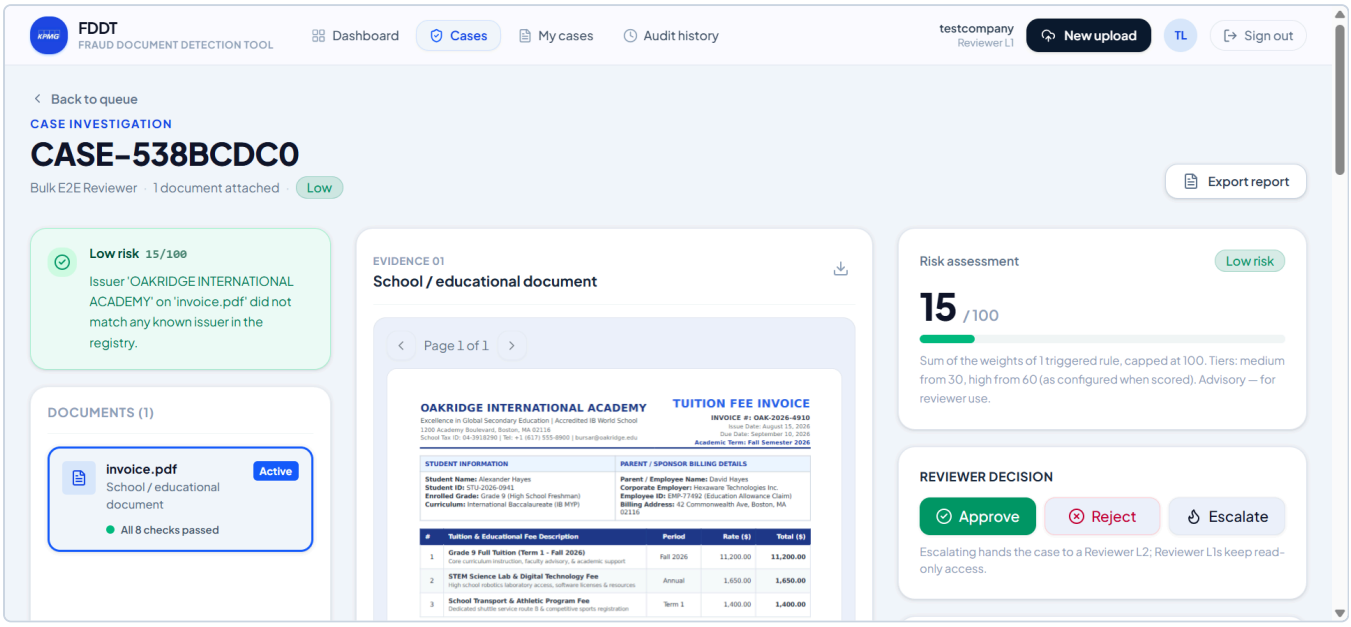
Review Queue

- **Risk Tier Badges**:
  - **HIGH RISK** (Score 70–100): Critical anomalies detected (e.g., severe metadata tampering, copy-move duplication, unrecognized issuer).
  - **MEDIUM RISK** (Score 30–69): Discrepancies or warnings (e.g., date mismatches, mathematical errors, missing expected fields).
  - **LOW RISK** (Score 0–29): Clear document with verified issuer and consistent fields.

- **Filters & Search**: Filter by Risk Tier, Case Type, Review Status, or keyword.

Full Case Analysis & Risk Breakdown

Opening any case from the queue reveals the complete **Case Detail** analysis:

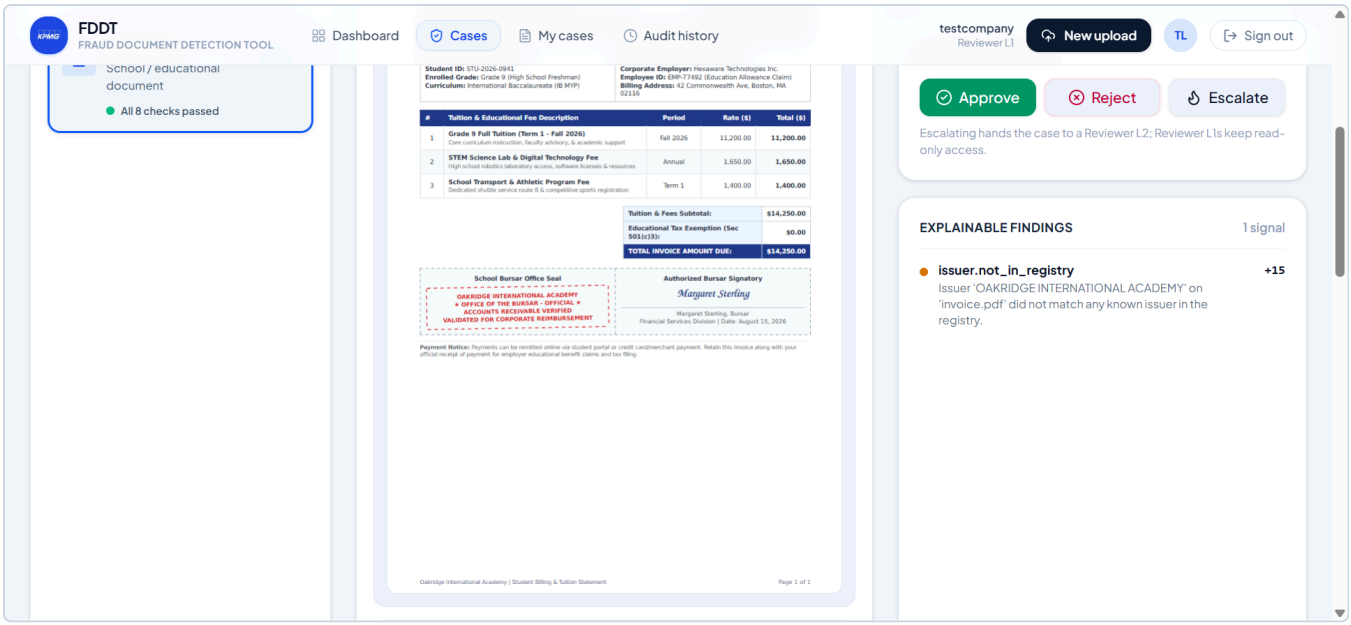


Reviewer Case Overview

1. **Risk Score Header**:
   - Prominently displays the composite Risk Score (0–100) and assigned Tier.
   - Shows case metadata: submitter email, case category, submission timestamp.
2. **Triggered Risk Reasons**:
   - Clear, explainable list of factors that contributed points to the score.
   - Example: "*Issuer not found in verified registry (+25 pts)*", "*PDF modified in image editor after creation (+20 pts)*".

Forensic & Automated Document Checks

Scroll down on the Case Detail page to inspect the comprehensive **Checks & Forensics Panel**:



Automated Checks & Forensics Panel

Every uploaded document undergoes an exhaustive battery of independent tests:

Check Name	What It Inspects
Issuer Verification	Cross-references extracted vendor name, tax ID, and address against your company's trusted Issuer Registry.
Field Validation	Verifies arithmetic totals (subtotal + tax = total), valid date formats, future dates, and required fields.
Cross-Document Check	Validates consistency between multiple documents in the same case (e.g., matching invoice amount with purchase order).
Metadata Forensics	Inspects PDF creation software, modification history, suspicious editing tools, and timestamp inconsistencies.
Copy-Move Forgery	Detects cloned regions (content copied and pasted from one part of the document to another).
Error Level Analysis (ELA)	Analyzes compression resave artifacts to identify spliced text or altered numbers.
Duplicate Detection	Computes cryptographic and perceptual hashes to detect duplicate submissions across historical cases.
Signature / Stamp Detection	Detects signature regions and compares them against reference samples for consistency.

Interactive PDF Visual Overlays

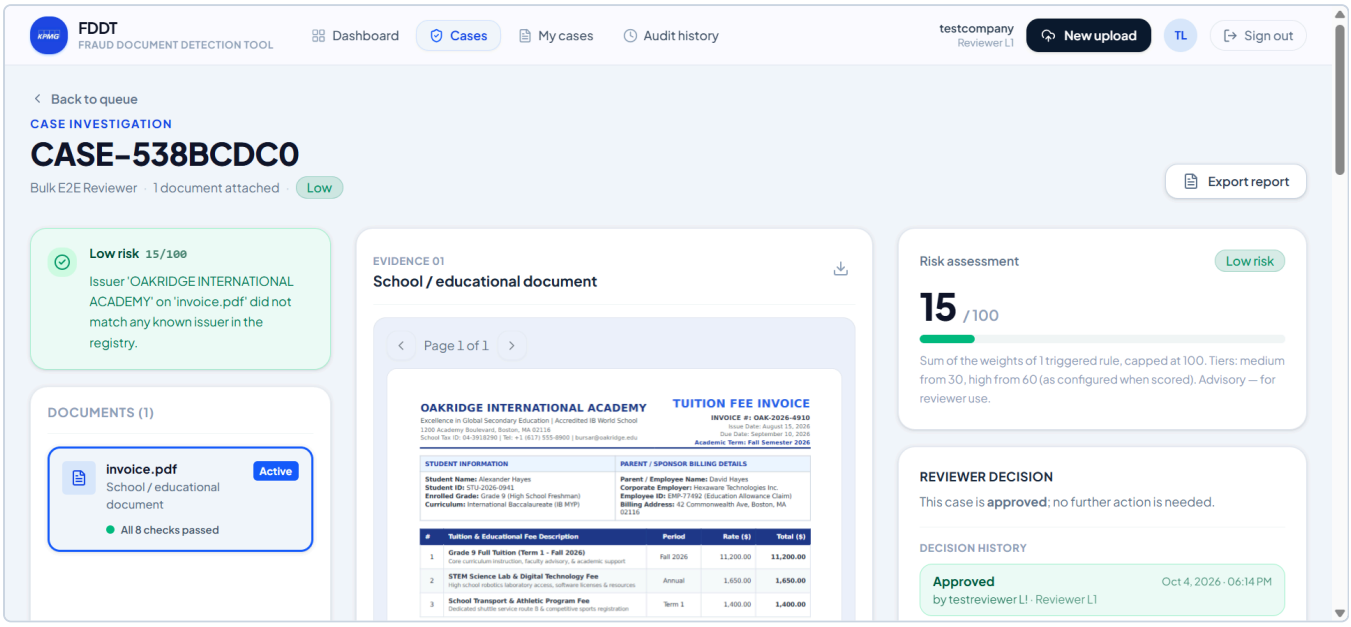
To visually inspect anomalies:

- Open the document preview within the Case Detail view.
- Color-coded bounding boxes highlight exact regions of concern:
  - **Red Box**: Tampering or Error Level Analysis anomaly.
  - **Orange Box**: Copy-move duplication source and target.
  - **Purple Box**: Arithmetic or field validation mismatch.
  - **Fuchsia Box**: Text set in a different font (or a second copy of the font) — likely typed in later.
  - **Teal Box**: Deleted or shortened text — the faint trace of erased text in a converted scan.

- **Blue Box**: Detected signature or official stamp.

Review Decisions (Approve, Reject, Escalate)

In the top-right header of the Case Detail screen, three decision actions are available:



Reviewer Decision Modal

1. **Approve**:
   - Use when all checks pass or minor flags have been verified as legitimate.
   - Enter optional approval notes and confirm. The case status updates to **Approved**.
2. **Reject**:
   - Use when fraudulent alteration or non-compliance is identified.
   - Select a mandatory **Rejection Reason** category and provide notes detailing the decision.
3. **Escalate**:
   - Use when a case is complex, high-value, or requires supervisor adjudication.
   - Adds an escalation note and routes the case directly to the **Reviewer L2** queue.

Exporting Case Forensic Reports

Click **Export report** at any time on the Case Detail page to generate an official, courtroom-ready **PDF Forensic Report**:

- Includes complete executive summary, breakdown of all automated checks, risk scoring explanation, audit trail, SHA-256 cryptographic document hashes, and high-resolution visual evidence pages with highlighted anomaly boxes.

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6. Reviewer L2 (Senior Reviewer / Supervisor) Guide

Reviewer L2 users have all analyst privileges plus senior decision authority and configuration control over their company's risk parameters.

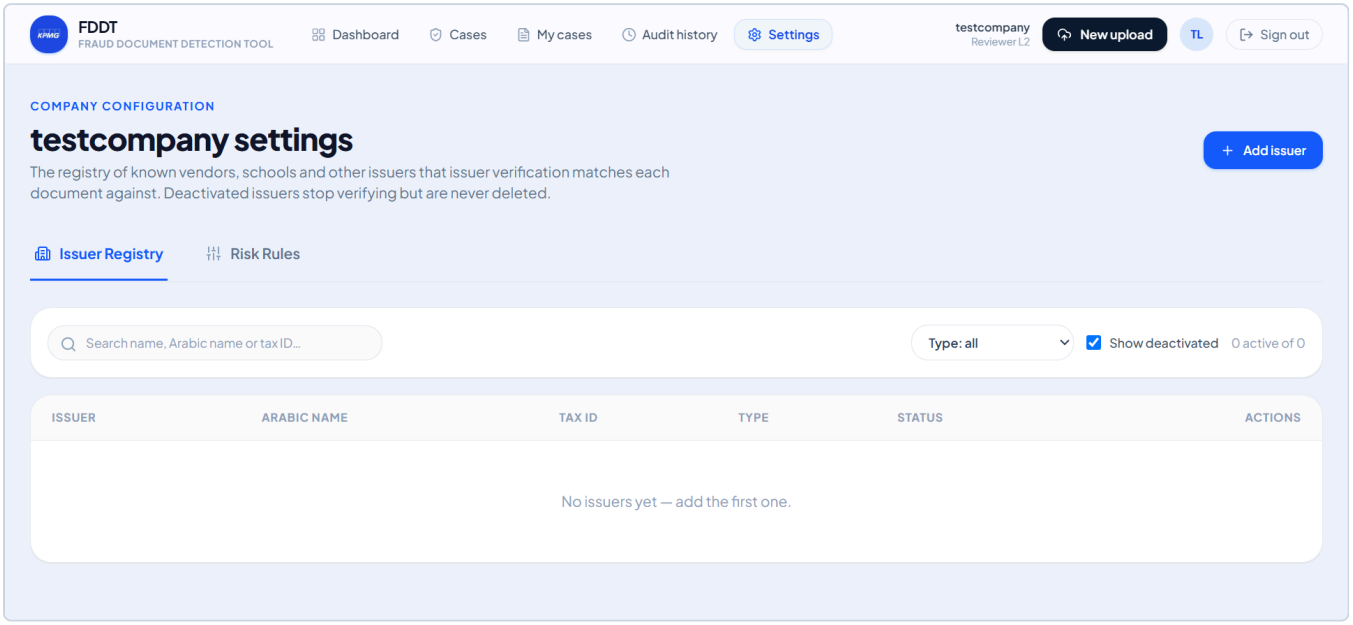
Managing Escalated Cases

Cases escalated by L1 analysts appear in the review queue with an **Escalated** status badge.

- Reviewer L2 users conduct deep reviews of these cases.
- Only Reviewer L2 users have the authority to grant final **Approval** or execute final **Rejection** on escalated files.

Company Issuer Registry

Under **Settings › Issuer Registry**, Reviewer L2 supervisors manage the company's verified vendors, schools, and partners:

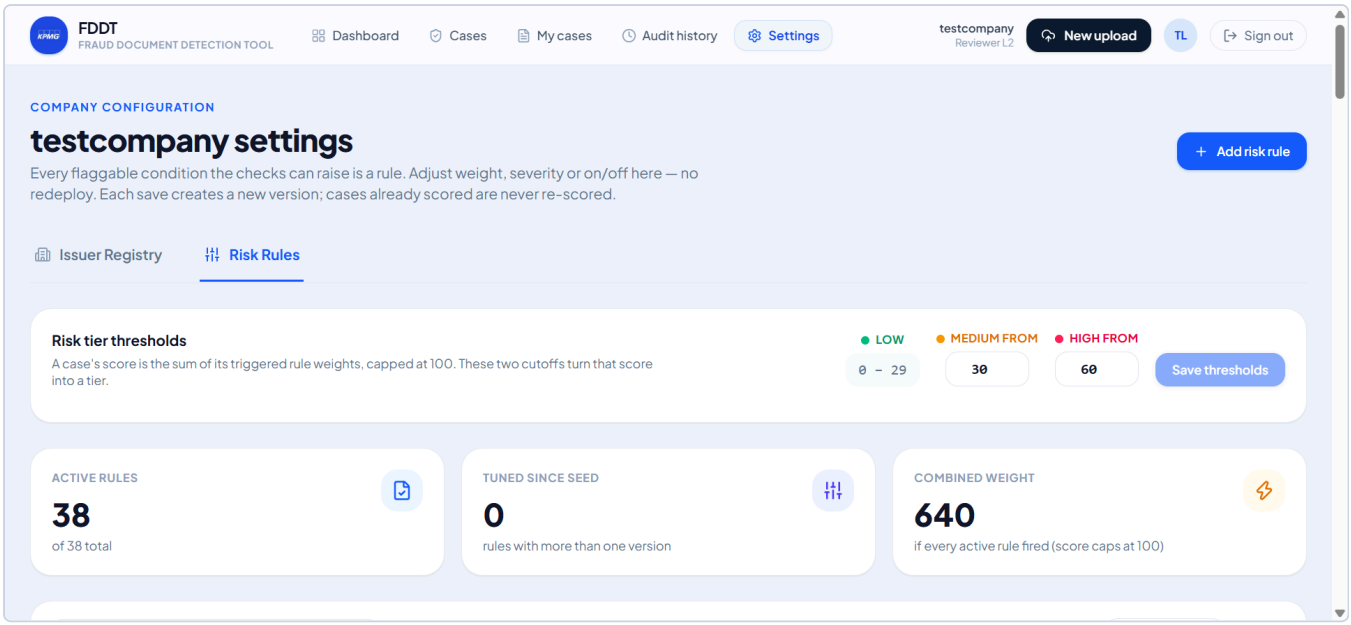


Issuer Registry Management

1. **Viewing Registry**: Lists all active and inactive registered issuers.
2. **Adding an Issuer**:
   - Click **Add Issuer**.
   - Enter Organization Name, Tax ID / Business Registration Number, official address, and known email domains.
   - Save the record. Incoming documents matching these details will pass Issuer Verification automatically.
3. **Deactivating / Reactivating**:
   - Deactivate obsolete vendors or reactivate previously deactivated entities with a single click.

Company Risk Rules & Thresholds

Under **Settings › Risk Rules**, Reviewer L2 supervisors customize how risk scores are calculated for their organization:



Company Risk Rules Management

1. **Adjusting Risk Rules**:
   - Toggle individual fraud detection checks on or off based on your business requirements.
   - Adjust point weights for specific findings (e.g., increasing penalty points for metadata anomalies).
2. **Configuring Risk Thresholds**:
   - Customize score boundaries for **Low**, **Medium**, and **High** risk classifications to match your company's risk tolerance.

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7. Getting Help & Administrative Requests

To maintain company security and data integrity, certain platform-level configurations are managed centrally.

> **Need assistance?**
>
> - To create new user accounts or adjust user roles
> - To reset an account password
> - To request an increase in maximum document upload size limits
> - To update your company profile or contact details
>
> **Please contact or email your administrator at `admin@example.com` (or your company's designated IT administrator) for this.**